Nonlinear population estimation: feedback gains, depth requirements, and a tested heuristic

Research checkpoint in adaptive threshold estimation

27 October 2026

Abstract

We investigate acquisition feedback as a possible escape from the verified single-coin and additive obstructions. In a two-gap experiment with a deterministic cap of two flips, an explicit feedback policy has smaller minimax squared error over the full global-rarity class than every preselected independent acquisition design, including arbitrary nonlinear postprocessing. Exact rational inequalities certify the separation. Conversely, polynomial moment matching gives two globally rare quadratic-law populations that are indistinguishable to every feedback algorithm with a fixed local observation cap. The same construction imposes a quantitative depth-tail requirement on unrestricted algorithms. A separate approximation theorem reduces the unrestricted expected-budget value to finite convex programs with explicit risk and budget errors for arbitrary known laws. A finite-prior greedy feedback estimator was implemented and evaluated in a frozen synthetic suite of7,680 runs. Its point estimates are encouraging, but establish neither a uniform error guarantee nor stable practical superiority. The arbitrary-law expected-budget minimax problem remains unresolved.

Evidence and scope. The mathematical statements are **PROVED-ON-PAPER** with separate adversarial agent review and explicitly limited exact arithmetic checks. They are not Lean formalized or externally peer reviewed. The simulation is **NUMERICAL**, in float64 with recorded seeds. No novelty or priority claim is made. A deterministic two-flip cap must not be replaced by an expected-work budget of two; an algorithm can satisfy the latter while occasionally taking more observations.

1 Experiment and unresolved objective

A fresh iid coin has hidden $D \sim \mu$ on $(0, 1/2]$ and $S \mid D = d \sim \mathrm{Bernoulli}(q(d))$. Its conditional flips are iid Bernoulli with mean $1/2 + (2S - 1)D$. The unknown measurable allocation satisfies $0 \le q \le 1$ and the global constraint $\theta(q) := \int q \, d\mu \le \rho$. Known μ, ρ do not reveal any sampled magnitude or sign. The algorithm sees only coin identities and bits and can create, revisit, interleave, abandon and stop adaptively. Genuine flips, including pilots, cost one each; deterministic computation and independent randomization are free in the statistical model. Actual computation time is accounted for separately.

The primary risk is minimax RMS under $\sup_q \mathbb{E}_q T \le B$. The candidate arbitrary-law formula is

$$\Psi_{\mu,\rho}(B^{-1/2}), \qquad \Psi_{\mu,\rho}(h) = \sup_{0 \le w \le 1, \int w \, d\mu \le \rho} \int \min\{w(d), h\sqrt{w(d)}/d\} \, \mu(dd).$$

For the exact quadratic law on $(0, 1/4]$, the candidate unsaturated order is $\sqrt{\rho L/B}$, $L = 1 + \log(1/\rho)$. The inherited single-coin and specified additive classes lose $\sqrt{1 + \log L}$. This pass asks whether genuine acquisition feedback can avoid that obstruction, without assuming that it does.

2 Feedback strictly improves a two-flip minimax subproblem

PROVED-ON-PAPER with certified finite rational arithmetic. This is a counterexample to removing cross-coin acquisition feedback without loss. It is not a separation under expected budgets, not an asymptotic rate separation, and not a lower bound for the full class $\mathcal{Q}_\rho$. No novelty or formalization claim is made.

Fix $\rho = 1/4$ and the known magnitude law $\mu = \frac{1}{2}\delta_{1/4} + \frac{1}{2}\delta_{1/8}$. The magnitude of each actual coin remains hidden. Consider the admissible two-point parameter class $\mathcal{H} = \{q_0 \equiv 0, q_1 \equiv 1/4\}$, with targets $\theta_0 = 0$ and $\theta_1 = 1/4$. All genuine coins are iid and every requested flip is charged. The following comparison imposes a *deterministic total cap of two flips*.

The noninteractive comparison class preselects its independent local coin policies using an arbitrary shared seed, independent of all coin tapes. Each local policy may stop using its own observations and the seed, but its acquisition choices cannot depend on any other coin's observations. The terminal estimator may be an arbitrary nonlinear function of all acquired data and the seed. The sum of the preselected policies' actual flips must obey the cap almost surely at every parameter; an external budget monitor that cancels one policy because of another's observations is cross-coin feedback and is outside this comparison class.

Theorem 2.1. *For this two-point problem, the minimax squared error of the noninteractive comparison class is strictly greater than* $15313/10^6$. *There is a feedback procedure with exactly two flips and maximum squared error strictly less than* $7623/500000$. *In particular its minimax squared-error advantage exceeds* $67/10^6$.

Likelihoods and the two fixed schedules. For a word h of length n with k heads, its one-coin probability under hypothesis j is the rational number

$$f_j(h) = \frac{1}{2} \sum_{d \in \{1/4, 1/8\}} \left[(1 - j/4)(1/2 - d)^k (1/2 + d)^{n-k} + (j/4)(1/2 + d)^k (1/2 - d)^{n-k} \right].$$

If two bits (x, y) come from the same coin, their probability is $p_{j,O}(x, y) = f_j(xy)$. For two fresh coins it is $p_{j,N}(x, y) = f_j(x) f_j(y)$. These probabilities use only actual observed bits, not conditional gap information supplied to the estimator.

Every member of the stated noninteractive class is statistically dominated by a random mixture of these two full schedules. To see this, condition on all independent seeds, including local randomization. If a local policy can request two flips with positive probability under q_0, no other policy can request even one with positive probability under q_0; otherwise independence would make total work greater than two with positive probability. If no policy can request two, at most two policies can request one. Finite words have strictly positive probability under both hypotheses, so the same statement holds for both hypotheses and for every potentially queried finite history. Unused pre-created identities reveal no data and can be omitted. A one-coin policy that stops early is dominated by completing two flips of that coin; policies with at most one flip each are dominated by observing one flip from each of two fresh coins. The original estimator can ignore the additional bits. This also handles zero observations. The chosen dominating schedule depends only on the seed, not on the hypothesis. Randomization therefore cannot lower its Bayes risk below the smaller Bayes risk of the two schedules.

A feedback policy and its exact estimator. Flip one fresh coin. After a tail, use the second flip on a fresh coin. After a head, revisit the original coin. In symbols, its probability of (x, y) is

$$p_{j,A}(x, y) = \begin{cases} f_j(x) f_j(y), & x = 0, \\ f_j(xy), & x = 1. \end{cases}$$

This costs exactly two flips in both hypotheses. Fix the rational number $\pi = 1075683009/2147483648$, interpreted only as a design constant, and output the posterior mean

$$\widehat{\theta}_A(x, y) = \frac{1}{4} \frac{\pi p_{1,A}(x, y)}{(1 - \pi) p_{0,A}(x, y) + \pi p_{1,A}(x, y)}.$$

The estimator is fully specified and takes values in $(0, 1/4)$; there is no uncharged pilot, latent-gap query, or unknown-parameter calibration.

Finite rational verification. For any schedule $P \in \{O, N, A\}$ the optimal Bayes squared error under prior π is exactly

$$\mathcal{B}_P(\pi) = \frac{1}{16}\pi(1-\pi) \sum_{x,y\in\{0,1\}} \frac{p_{0,P}(x,y)p_{1,P}(x,y)}{(1-\pi)p_{0,P}(x,y) + \pi p_{1,P}(x,y)}.$$

Let $r_{j,A} = \sum_{x,y} p_{j,A}(x,y)(\widehat{\theta}_A(x,y) - j/4)^2$. The script `feedback_certificate.py` computes all these fractions exactly, checks that each likelihood vector sums to one, and checks the independent posterior-variance identity for every Bayes risk. It verifies

$$\max_j r_{j,A} < \frac{7623}{500000} < \frac{15313}{10^6} < \min\{\mathcal{B}_O(\pi), \mathcal{B}_N(\pi)\}.$$

For orientation only, the adaptive maximum is approximately 0.01524525010158912, while the smaller fixed-schedule Bayes lower bound is approximately 0.015313875487111883. The inequalities, rather than these decimal displays, are the certificate. Every minimax risk is at least its Bayes risk under the specified prior, so the displayed chain proves Theorem 2.1.

The script also gives exact certificates for magnitude laws with equal weights at $(1/4, 1/64)$ and $(1/4, 1/1024)$, with separately pinned dyadic design priors. Thus strict feedback gain is not specific to neighboring gap scales. It remains a finite cap-two effect: none of the certificates establishes a growing advantage, a critical-law lower bound, or the same comparison for algorithms having only a uniform expected-work budget.

Why this matters, and the remaining gap. An unrestricted estimator cannot in general be replaced at no loss by a preselected independent acquisition design and an optimal nonlinear postprocessor. The first bit changes whether information should be acquired from the same latent coin or a fresh one, even in this tiny admissible subproblem. The certificate does not imply that feedback defeats the critical logarithmic obstruction in the worst case over arbitrary q. Amplifying the finite advantage into such a statement requires a hard family or construction whose advantage grows with the number of gap scales, while controlling the original uniform expected budget. No such amplification is proved here.

3 A strict feedback advantage over the full rarity class

PROVED-ON-PAPER with exact rational verification. This strengthens the preceding two-profile example to the entire global-rarity class for one finite-support magnitude law. It retains the deterministic total cap of two flips; no expected-budget comparison or asymptotic rate separation is asserted. The earlier certificate is preserved.

Let $\rho = 1/4$ and $\mu = \frac{1}{2}\delta_{1/4} + \frac{1}{2}\delta_{1/8}$. Write $q_1 = q(1/4)$, $q_2 = q(1/8)$. The full admissible class is exactly $q_1, q_2 \geq 0$, $q_1 + q_2 \leq 1/2$; the individual upper bounds by one are automatic. Put

$$\theta = \tfrac{1}{2}(q_1 + q_2), \qquad A = \mathbb{E}[Dq(D)] = q_1/8 + q_2/16.$$

The feasible (A, θ) region is the triangle with vertices

$$(0,0), \quad (1/32, 1/4), \quad (1/16, 1/4).$$

Equivalently $0 \leq A \leq 1/16$ and $4A \leq \theta \leq \min\{8A, 1/4\}$.

The noninteractive comparison class is exactly the one defined in the preceding theorem: preselected independent local acquisition policies, arbitrary shared and local seeds, and arbitrary nonlinear terminal postprocessing, with a deterministic total cap of two flips and no cross-coin acquisition feedback or outcome-dependent cancellation.

Theorem 3.1. *Over the entire class $\mathcal{Q}_{1/4}(\mu)$, the minimax MSE of the noninteractive comparison class is strictly greater than $15475/10^6$. An explicit two-flip feedback estimator has worst-case MSE strictly less than $15443/10^6$. Its minimax squared-error advantage therefore exceeds $32/10^6$.*

Proof. One fresh bit has head probability $p = 5/16 + 2A$. In the word order (TT, TH, HT, HH), the same-coin probabilities are

$$P_O(A) = (61/128 - 2A,\ 27/128,\ 27/128,\ 13/128 + 2A),$$

whereas the two-fresh probabilities are $P_N(A) = ((1-p)^2, (1-p)p, p(1-p), p^2)$. These formulas integrate out the hidden retained magnitude and sign; the algorithm observes neither. In particular the two-flip laws depend on A, while the target θ still varies over its feasible interval.

The feedback policy takes a fresh coin after the first tail and revisits after the first head. Its four probabilities are

$$P_A(A) = ((11/16 - 2A)^2,\ (11/16 - 2A)(5/16 + 2A),\ 27/128,\ 13/128 + 2A).$$

It takes exactly two charged flips. Define the fixed rational constant $\pi = 1074499095/2147483648$, and set $a_x = P_A(0)_x$, $b_x = P_A(1/32)_x$ for its four outcomes. Its output table is

$$y_x = \frac{1}{4}\frac{\pi b_x}{(1-\pi)a_x + \pi b_x}.$$

In the displayed word order this is exactly

$$(y_{TT}, y_{TH}, y_{HT}, y_{HH}) = \left(\frac{26862477375}{237281040413}, \frac{3223497285}{24696819223}, \frac{1074499095}{8589934592}, \frac{22564480995}{146053120736}\right).$$

Although this table was constructed from the null and the admissible hard alternative $(q_1, q_2) = (0, 1/2)$, the same fixed table is used for every $q \in \mathcal{Q}_{1/4}(\mu)$. It requires no unknown-parameter calibration.

Its risk is the polynomial $R(A, \theta) = \sum_x P_A(A)_x (y_x - \theta)^2$. For fixed A, its second derivative in θ is exactly two, so the maximum over the vertical feasible interval is at an endpoint. It remains to maximize on the three edges $\theta = 4A$ for $A \in [0, 1/16]$, $\theta = 8A$ for $A \in [0, 1/32]$, and $\theta = 1/4$ for $A \in [1/32, 1/16]$. On each diagonal edge the risk has degree at most three in A, and on the horizontal edge degree at most two. The certificate computes their exact rational coefficients and verifies that every edge's second derivative exceeds respectively 30, 124, and $3/100$ at both endpoints. Since each such second derivative is affine, it is positive throughout the edge. Thus each edge is convex and the maximum over the entire triangle occurs at one of its three vertices. This is a polynomial proof, not inference from a numerical grid.

The largest of the three exact vertex risks is less than $15443/10^6$; its approximate value is 0.015442586588172843. This proves the full-class upper bound.

For the lower bound, use only the null $(0, 0)$ and hard alternative $(0, 1/2)$, with probabilities $1 - \pi, \pi$. Both belong to the full class. For either fixed schedule $P \in \{O, N\}$ its optimal Bayes risk is

$$\mathcal{B}_P = \frac{1}{16}\pi(1-\pi)\sum_x \frac{P(0)_x P(1/32)_x}{(1-\pi)P(0)_x + \pi P(1/32)_x}.$$

The preceding dominance argument for noninteractive cap-two designs applies unchanged: the null has full finite-word support, so conditioning on all seeds permits either one local policy with two flips or at most two local policies with one flip each. Completing any early stops gives one of these schedules and cannot increase the optimal Bayes risk. The comparison permits arbitrary nonlinear estimates; the posterior mean used in the Bayes formula is optimal among all of them. Therefore every noninteractive procedure's worst-case MSE over the full class is at least $\min\{\mathcal{B}_O, \mathcal{B}_N\}$.

The exact certificate verifies

$$\max_{\text{three vertices}} R < \frac{15443}{10^6} < \frac{15475}{10^6} < \min\{\mathcal{B}_O, \mathcal{B}_N\}.$$

The smaller Bayes lower bound is approximately 0.015475854685963823. This proves the strict separation. □

All probabilities, coefficients, outputs and inequalities are recorded in `full_class_certificate.json`, generated by `full_class_certificate.py` using rational arithmetic only. The design prior was selected by a bounded exploratory rational bisection; its optimality is not asserted or used. The independent proof of the full triangle maximum is the essential new certification step.

This result shows that acquisition feedback can improve minimax performance even when the adversary ranges over *all* globally rare sign allocations for a known magnitude law. It does not show that the improvement grows with rarity or gap-scale count. The comparison for a uniform expected budget remains separate, since a noninteractive procedure allowed occasional work above two is not covered by the cap-two dominance argument.

4 The feedback effect for arbitrary hidden magnitude laws

PROVED-ON-PAPER. This finite two-flip information comparison concerns the submodel $q_t(d) \equiv t$. It is not a minimax comparison over arbitrary q or an asymptotic normality assertion.

Let μ be any law on $(0, 1/4]$. Write $m = \mathbb{E}D > 0$, $v = \mathbb{E}D^2$, and

$$p_H = \tfrac{1}{2} - m, \quad p_T = \tfrac{1}{2} + m, \quad h = \tfrac{1}{4} - m + v, \quad c = \tfrac{1}{4} - v, \quad z = \tfrac{1}{4} + m + v.$$

These positive quantities are the first-head/first-tail probabilities and the word probabilities of HH, HT, TT at $t = 0$. Their derivatives along $q_t = t$ are $2m, -2m$ for H, T and $2m, 0, -2m$ for HH, HT, TT. The derivatives are from the admissible right side at $t = 0$; the finite word probabilities are positive and affine, so their information is defined.

Proposition 4.1. *The information J from a fresh bit and conditional information J_H, J_T from revisiting after a head or tail satisfy*

$$J = \frac{4m^2}{p_H p_T}, \qquad J_H = \frac{4m^2 c}{p_H^2 h}, \qquad J_T = \frac{4m^2 c}{p_T^2 z}, \qquad J_T < J < J_H.$$

The feedback policy A (fresh after a tail, revisit after a head) therefore has strictly more information than either fixed two-flip schedule:

$$I_A = J + p_H J_H + p_T J, \quad I_O = J + p_H J_H + p_T J_T, \quad I_N = 2J, \qquad I_A > \max\{I_O, I_N\}.$$

Proof. The Bernoulli information formula gives J. After a head the next-head probability is h/p_H, with derivative $2mc/p_H^2$; after a tail it is c/p_T, with derivative $2mc/p_T^2$. Applying the same formula gives J_H, J_T. Since $cp_H < p_T z$, we have $J_T < J$. Direct algebra gives $cp_T - p_H h = m - m^2 - v$. The support restriction implies $m^2 \le m/4$ and $v \le m/4$, so this expression is at least $m/2 > 0$, proving $J_H > J$. The second bit's score increment has conditional mean zero, so its information adds to that of the first. This proves the remaining identities. □

For the exact quadratic law with $a = 1/4$, $m = 1/6$, $v = 1/32$, and

$$J = \frac{1}{2}, \quad J_H = \frac{21}{11}, \quad J_T = \frac{21}{172}, \qquad I_A = \frac{97}{66}, \quad I_O = \frac{576}{473}, \quad I_N = 1.$$

Thus feedback changes information even for the critical law. All compared procedures use two genuine flips and no realized-gap information.

What does not amplify automatically. For hypotheses $q = 0$ and $q = \rho$ with equal prior, any of these three fixed finite experiments has Bayes risk, for fixed μ,

$$\mathcal{B}_P(\rho) = \frac{\rho^2}{4} - \frac{\rho^4}{16} I_P(0) + O_\mu(\rho^5).$$

To prove this, write its finitely many positive cell probabilities as $p_\rho(x) = p_0(x) + \rho g(x) + O(\rho^2)$ and expand the exact identity $\mathcal{B}_P = (\rho^2/2)\sum_x p_0(x)p_\rho(x)/(p_0(x) + p_\rho(x))$. Normalization cancels the linear term and the sum of second derivatives of the probabilities. The remaining second-order term is $-\rho^4 \sum_x g(x)^2/p_0(x)/16$. The positive finite denominators bound the Taylor remainder. The feedback schedule is fixed and its finite cell probabilities are polynomial, so this also applies to it.

Hence its equal-prior Bayes advantage persists for sufficiently small ρ for every fixed μ, but is of order ρ^4 against a leading risk of order ρ^2. This does not give a growing minimax advantage. Repeating two-flip batches would also restrict the comparison experiment: it does not bound competitors that revisit more deeply, use other feedback, or have only an expected-work constraint. The missing multiscale argument cannot be replaced by multiplying this finite advantage.

5 A limit on shallow calibration, including feedback

Evidence and limits. This is a paper proof, accompanied by exact rational finite identity checks. It is not Lean formalized or externally peer reviewed. No novelty claim is made: the argument uses polynomial moment matching. It rules out a bounded local-depth pilot strategy, including arbitrary feedback between coins. It does not resolve the unrestricted rare-event minimax problem.

Experiment. Fix $0 < a < 1/2$ and the known law $\mu(dd) = 2d\,dd/a^2$ on $(0, a]$. The sign allocation q is unknown, $0 \le q \le 1$ and $\int q\,d\mu \le \rho$, where $0 < \rho \le 1/4$ is known. Each iid fresh coin has hidden $D \sim \mu$, hidden $S \mid D = d \sim \mathrm{Ber}(q(d))$, and supplies conditionally iid $\mathrm{Ber}(1/2 + (2S-1)D)$ observations. An algorithm may use arbitrary independent randomization, create, revisit, interleave and abandon coins, and let all these decisions depend on all earlier observations. Only flips and identities are observed. Every flip is charged. The restriction considered below is solely that each individual coin supplies at most $2K$ flips.

Lemma 5.1 (An elementary bounded polynomial)**.** *For integer $K \ge 1$, let P_n be the degree-n Legendre polynomial defined by Rodrigues' formula, $n = 2K+1$, and put*

$$c = P_n'(0), \qquad u_K(x) = P_n(x)/(cx), \quad u_K(0) = 1.$$

Then u_K is an even polynomial of degree $2K$ satisfying

$$|u_K(x)| \le 1 \quad (0 \le x \le 1), \qquad \int_0^1 x^{2j+2} u_K(x)\,dx = 0 \quad (0 \le j < K),$$

and

$$\int_0^1 2x u_K(x)\,dx = \frac{1}{(K+1)(2K+1)} =: m_K.$$

Proof. For completeness the polynomial facts needed here follow from

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n.$$

Expansion gives

$$c = (-1)^K(2K+1)\binom{2K}{K}4^{-K}, \qquad P_{2K}(0) = (-1)^K\binom{2K}{K}4^{-K} =: v_K.$$

It also gives the differential and derivative identities

$$(1-x^2)P_n'' - 2xP_n' + n(n+1)P_n = 0, \qquad P_{n+1}' - P_{n-1}' = (2n+1)P_n.$$

These identities can equivalently be verified coefficient by coefficient from the displayed expansion. Rodrigues' formula at $x = 1$ gives $P_j(1) = 1$: in the derivative of $(x-1)^j(x+1)^j$ only the term differentiating the first factor j times survives at 1.

Substitute $P_n = cxu_K$ in the differential equation and set $\lambda = n(n+1) - 2 = 2K(2K+3) > 0$. On $(0,1)$,

$$(1-x^2)u_K'' + (2/x - 4x)u_K' + \lambda u_K = 0.$$

The energy $H(x) = u_K(x)^2 + (1-x^2)u_K'(x)^2/\lambda$ satisfies

$$H'(x) = (6x - 4/x)u_K'(x)^2/\lambda.$$

Thus H decreases up to $\sqrt{2/3}$ and increases thereafter, and its maximum on $[0,1]$ is at an endpoint. Since u_K is even, $H(0) = 1$; $H(1) = 1/c^2 \le 1$. The last inequality follows from $|c| \ge 1$: the value at $K = 0$ is 1 and the ratio of successive absolute values is $(2K+3)/(2K+2) > 1$. Since $H \ge u_K^2$, the asserted bound follows.

Integrating Rodrigues' formula by parts n times shows that P_n is orthogonal on $[-1,1]$ to every polynomial of degree less than n; the boundary terms vanish because $(x^2-1)^n$ has zeros of order n at both endpoints. Therefore

$$\int_0^1 x^{2j+2} u_K(x)\,dx = \frac{1}{2c}\int_{-1}^1 x^{2j+1} P_n(x)\,dx = 0.$$

Finally $v_{K+1} = -(2K+1)v_K/(2K+2)$ and the derivative identity give

$$\int_0^1 P_{2K+1}(x)\,dx = \frac{v_K - v_{K+1}}{4K+3} = \frac{v_K}{2K+2}.$$

Multiply by $2/c$, where $c = (2K+1)v_K$, to obtain m_K. □

Theorem 5.2 (Feedback does not repair bounded-depth nonidentifiability)**.** *For any $K \ge 1$, define $h = a\sqrt{\rho}$ and*

$$q_\pm(d) = \frac{1 \pm u_K(d/h)}{2}\,\mathbf{1}_{\{d \le h\}}.$$

Both are admissible allocations, with

$$\theta_\pm = \frac{\rho}{2}(1 \pm m_K), \qquad \theta_+ - \theta_- = \frac{\rho}{(K+1)(2K+1)}.$$

Every population algorithm obeying the per-coin cap $2K$ has identical finite-prefix transcript distributions under these two allocations, including arbitrary cross-coin feedback and pilot-dependent acquisition. If its terminal output is measurable and the procedure terminates almost surely under both laws, its output distributions also coincide. Consequently its worst-case RMS error is at least

$$\frac{\rho}{2(K+1)(2K+1)},$$

regardless of its expected total work or number of coins.

Proof. Lemma 5.1 implies $0 \le q_\pm \le 1$. A change of variable $d = hx$ gives the displayed targets, each at most ρ. This uses only the global rarity constraint: the pointwise allocations can be as large as one in the small difficult population.

For a particular bit word w of length r and head count s, its likelihood conditional on a positive or negative sign is

$$\ell_\pm(d) = (1/2 \pm d)^s (1/2 \mp d)^{r-s}.$$

The difference $\ell_+(d) - \ell_-(d)$ is an odd polynomial of degree at most r. When $r \le 2K$, all its monomials have degrees $2j+1$ with $j < K$. For each such degree,

$$\int (q_+ - q_-)(d) d^{2j+1}\,\mu(dd) = \frac{2h^{2j+3}}{a^2}\int_0^1 u_K(x) x^{2j+2}\,dx = 0.$$

Hence the probability of every local word of length at most $2K$ is exactly the same under the two allocations.

Condition on the algorithm's independent seed. For any finite global transcript, its probability is the product of the probabilities of the local words of the distinct coins it sampled, multiplied by the indicator that its scheduling and stopping decisions agree with the transcript. The indicator is parameter independent; every local word has length at most $2K$. Thus each finite prefix has the same probability under the two laws. Integrating the seed preserves equality. Coins may be indexed by their first charged observation, so free unused identities cause no difficulty. Equivalently, equality follows successively from identical conditional next-bit probabilities for every eligible local history.

For an almost surely terminating procedure, a terminal event intersected with the event of termination within N observations belongs to a finite prefix. Its probabilities coincide for every N and converge to the full terminal probabilities under both laws. Finally, if Z has this common output law and $\Delta = \theta_+ - \theta_-$, then pointwise

$$\tfrac{1}{2}[(Z-\theta_+)^2 + (Z-\theta_-)^2] = (Z - (\theta_+ + \theta_-)/2)^2 + \Delta^2/4 \geq \Delta^2/4.$$

Taking expectations, including infinite second moments, proves the RMS bound. □

Theorem 5.3 (A depth-tail necessity for unrestricted feedback)**.** *Keep the two allocations and target gap $\Delta_K = \rho m_K$ from Theorem 5.2. Now allow an unrestricted population algorithm, including arbitrarily many observations on a coin, and assume almost-sure termination under these two laws. Let E_K be the event that some individual coin receives more than $2K$ flips. Then*

$$p_K := \mathbb{P}_+(E_K) = \mathbb{P}_-(E_K).$$

If the RMS error is at most ε under both laws, then

$$p_K \geq 1 - \frac{4\varepsilon^2}{\Delta_K^2}.$$

In particular, $\Delta_K \geq 4\varepsilon$ forces $p_K \geq 3/4$ under each law. Moreover, separately under both laws,

$$\mathbb{E}_\pm T \geq (2K+1)p_K.$$

The right side of the probability lower bound is informative only when positive; it can always be replaced by its maximum with zero.

Proof. Construct a capped experiment from the given algorithm. Follow all its decisions until it terminates, or until it is about to request the $(2K+1)$st observation of some coin. In the latter case stop immediately *before* that observation and record an exit flag. Otherwise record the original terminal output and a no-exit flag. This auxiliary experiment obeys the local cap $2K$. It terminates almost surely under both laws: on paths with no exit it follows the original terminating procedure, and on exit paths it stops earlier. Theorem 5.2 therefore gives equality of its terminal laws. In particular the exit probabilities agree, proving the first assertion. This equality concerns the event of the first excessive query, whose decision is made using only earlier allowed observations; it does not assert equality of later unrestricted transcripts.

Let ν be the subprobability measure of the original output together with the no-exit event. The capped experiment shows that it is the same under both laws and has total mass $1 - p_K$. If Z denotes the output coordinate, nonnegativity of squared errors gives

$$\begin{aligned}\mathbb{E}_+(Z-\theta_+)^2 + \mathbb{E}_-(Z-\theta_-)^2 &\geq \int \left[(z-\theta_+)^2 + (z-\theta_-)^2\right] \nu(dz) \\ &\geq (1-p_K)\Delta_K^2/2.\end{aligned}$$

The assumed RMS bounds make the left side at most $2\varepsilon^2$, giving the probability claim. This common-submeasure argument also gives total variation at most p_K between the unrestricted output laws; there is no factor two, since their common part has mass $1 - p_K$. On the exit event the original algorithm actually requests and obtains a coin's $(2K+1)$st flip, so its total charged work is at least $2K+1$. Taking expectations proves the work bound, even when the expectations are infinite. □

The first nontrivial calibration example. For $K = 2$,

$$u_2(x) = 1 - \frac{14}{3}x^2 + \frac{21}{5}x^4, \quad m_2 = 1/15,$$

and the targets are $8\rho/15$ and $7\rho/15$. No population policy with at most four flips per coin can have uniform RMS error below $\rho/30$, even with arbitrarily many pilot and production coins and nonlinear output. The two profiles agree on every pilot statistic measurable from these observations. A pilot that fits one of the profiles and uses that fit for later calibration therefore cannot validate its target uniformly.

What remains open. This theorem permits arbitrary feedback but imposes a fixed local cap. The main expected-budget experiment has no such cap. The depth-tail corollary nevertheless applies to that unrestricted experiment: sufficiently accurate procedures must enter greater local depths with a quantified probability. The target gap decreases quadratically with the cap and proves neither an iterated logarithm in the unrestricted rate nor the optimality of any proposed estimator. An attempt to turn pilot-learned survivor weights into a uniform estimator remains unsuccessful: calibration requires controlling the response-weighted error of the nuisance fit, and finite-depth agreement does not do that.

6 What clipping does and does not inherit from a variance lower bound

PROVED-ON-PAPER; elementary reduction, priority unresolved. This section was constructed by the audit agent and received a separate agent review recorded in `REVIEW.md`; it is not Lean formalized or externally peer reviewed. It gives a conditional transfer, not a theorem that every estimator in the original model satisfies its tail premise.

Write $\Pi_\rho(x) = \min\{\rho, \max\{0, x\}\}$ and assume the true parameter $\theta \in [0, \rho]$. The clipping interval is known.

Proposition 6.1 (A bounded-kurtosis transfer)**.** *Suppose F is unbiased for θ, with finite variance σ^2 and*

$$\mathbb{E}(F - \theta)^4 \le K\sigma^4, \qquad 1 \le K < \infty.$$

Then

$$\mathbb{E}(\Pi_\rho(F) - \theta)^2 \ge \frac{\min\{\sigma^2, \rho^2\}}{256K^2}.$$

This includes $\theta = 0$ and $\theta = \rho$.

Proof. The case $\sigma = 0$ is immediate. Put $X = F - \theta$. Holder's inequality gives

$$\sigma^2 \le (\mathbb{E}|X|)^{2/3}(\mathbb{E}X^4)^{1/3}, \quad \text{so} \quad \mathbb{E}|X| \ge \sigma/\sqrt{K}.$$

Since $\mathbb{E}X = 0$, both positive and negative parts have mean at least $\sigma/(2\sqrt{K})$. If $\theta \le \rho/2$ take $Y = X_+$; otherwise take $Y = X_-$. In both cases this is the error directed toward the more distant endpoint of the clipping interval. Write $m = \mathbb{E}Y$. From $\mathbb{E}Y^2 \le \sigma^2$ and Cauchy--Schwarz,

$$\mathbb{P}\{Y \ge m/2\} \ge \frac{m^2}{4\mathbb{E}Y^2} \ge \frac{1}{16K}.$$

On this event the clipped absolute error is at least $\min\{m/2, \rho/2\} \ge \min\{\sigma/(4\sqrt{K}), \rho/2\}$. Consequently its squared expectation is at least $\min\{\sigma^2/(256K^2), \rho^2/(64K)\}$, which implies the claim. □

If these premises hold for every q with one common K, taking suprema gives a clipped RMS lower bound of $(16K)^{-1}\min\{\rho, \sqrt{\sup_q \mathrm{Var}_q F}\}$. The inherited additive variance--work lower bound therefore survives clipping for unbiased additive estimators with uniformly bounded standardized fourth moment. The constant explicitly deteriorates with K; no such moment hypothesis was established for the inherited randomized telescopes. This proposition does not cover arbitrary biased raw estimators.

Proposition 6.2 (Why there is no unrestricted variance transfer)**.** *Let F_0 be any unbiased square-integrable raw estimator, and independently draw $Z = 0$ with probability $1 - \varepsilon$ and $Z = \pm M$ with probability $\varepsilon/2$ each. Set $F = F_0 + Z$. Then the acquisition cost is unchanged, F remains unbiased, and*

$$\operatorname{Var}_q F = \operatorname{Var}_q F_0 + \varepsilon M^2, \qquad \mathbb{E}_q(\Pi_\rho(F) - \theta)^2 \le \mathbb{E}_q(F_0 - \theta)^2 + \varepsilon\rho^2.$$

Proof. The variance identity uses independent mean-zero seed noise. On $Z = 0$, projection does not increase squared error. On the other event the clipped estimate and target both belong to $[0, \rho]$, so their distance is at most ρ. Taking expectations proves the bound. □

This construction applies within the original quadratic-law experiment by using its existing unbiased additive estimator for F_0. The noise is a square-integrable seed offset for each finite ε, M, so it also respects the inherited additive class. Taking $M = 1/\varepsilon$ makes the raw variance diverge while the clipping penalty tends to zero. This does not construct an estimator beating the additive minimax rate: it disproves an inference from a particular estimator's raw variance to its clipped risk. To make the clipped risk arbitrarily small, first select an increasingly accurate unbiased base estimator at its actual charged budget, and then take ε small. A fixed base estimator alone establishes variance divergence with a bounded clipping penalty, not vanishing clipped risk.

There is also a simple tail-energy transfer without unbiasedness. Let $D = F - \Pi_\rho(F)$. The triangle inequality in L^2 gives

$$\|\Pi_\rho(F) - \theta\|_2 \ge \|F - \theta\|_2 - \|D\|_2.$$

Thus a separately proved uniform bound on the clipping displacement can transfer a raw-risk lower bound. The displacement bound is an additional obligation, not a consequence of finite variance or expected work.

7 A faithful approximation of the unrestricted expected-budget problem

Evidence and purpose. These are complete written arguments, not Lean-checked or externally peer-reviewed results. No novelty claim is made. The result justifies a finite approximation route for the actual unrestricted minimax problem. It does not identify its rate or make a finite computation an exact certificate for an unbounded experiment.

Model and computational access. Fix a known probability law μ on $(0, 1/2]$ and known $0 < \rho \le 1/4$. A fresh coin draws $D \sim \mu$ and $S \mid D = d \sim \mathrm{Ber}(q(d))$, and supplies conditionally iid $\mathrm{Ber}(1/2 + (2S - 1)D)$ bits. The allocation q is measurable, $0 \le q \le 1$, and $\theta(q) = \int q\, d\mu \le \rho$. An algorithm sees bits and identities only, and may create, interleave, revisit, abandon and stop adaptively, with arbitrary independent seeds. Every actual bit is charged; deterministic computation is not charged.

For the mathematical approximation we grant exact values of the masses of the finitely many chosen bins, and exact real arithmetic on the resulting finite coefficients. This is an explicit known-law oracle convention. An arbitrary measure given without an effective representation need not have computable bin masses. An executable certified computation would need exact coefficients or separately certified coefficient errors; neither effective computability for every measure nor a finite-program solve is claimed here. The quadratic law with rational endpoints has exact rational bin masses. Knowing the law never reveals an individual coin's latent magnitude.

Write $V_\mu(B) = R_{\mu,\rho}(B)^2$ for minimax squared error at uniform expected budget $B > 0$. We may restrict all outputs to $[0, \rho]$: projection onto this interval decreases squared loss for every admissible target and does not change work. Let $V_{\mu,N}(B)$ impose also at most N total flips, where $N \ge 1$ is an integer. This is a total cap, not a per-coin cap.

Choose a finite interval partition of $(0, 1/2]$, retain its positive-mass bins $A_1, \dots, A_m$, and choose pairwise-distinct representatives $d_j \in (0, 1/2]$ such that $|d - d_j| \le \eta$ on A_j. Put $\pi_j = \mu(A_j)$ and $\mu_\eta = \sum_j \pi_j \delta_{d_j}$. Such partitions and distinct interior representatives exist for every $\eta > 0$; discarded bins

are μ-null and all allocation lifts may be defined arbitrarily on them. For $0 < \gamma \leq 1$, define the finite parameter set

$$Q_\gamma = \{q \in (\gamma\mathbb{Z}_{\geq 0})^m : 0 \leq q_j \leq 1,\ \sum_j \pi_j q_j \leq \rho\}.$$

Let $G_{N,\eta,\gamma}(B)$ be the minimax squared error on this finite parameter set and rounded magnitude law, using the full adaptive population experiment with at most N total flips and uniform expected work at most B. The actual representatives and bin masses, not merely η, are part of this notation. The finite experiment still hides every sampled type.

Theorem 7.1 (Explicit expected-budget approximation)**.** *Put*

$$C = N^2(\eta + \gamma), \qquad E = \rho^2 N(\eta + \gamma) + 2\rho\gamma.$$

Then

$$|V_\mu(B) - G_{N,\eta,\gamma}(B)| \leq \frac{\rho^2 B}{N} + E + \frac{\rho^2 C}{4(B + C)}. \tag{1}$$

In particular, for fixed $\mu, \rho, B > 0$, taking $N \to \infty$ and partitions with $\eta = N^{-3}$ and grids $\gamma = N^{-3}$ makes the right side tend to zero. Each approximating experiment has a finite-dimensional convex minimax formulation retaining all adaptive coin allocations and exactly the specified uniform expected-work constraint.

Lemma 7.2 (Truncation preserves the expected-budget comparison)**.**

$$V_\mu(B) \leq V_{\mu,N}(B) \leq V_\mu(B) + \rho^2 B/N.$$

Proof. The first inequality is inclusion of policy classes. For any nearly optimal expected-budget policy, follow it until it terminates or reaches N flips. If it has not terminated by that point, output $\rho/2$. Work only decreases. On $\{T \leq N\}$ its clipped output is unchanged; on $\{T > N\}$ the squared loss increase is at most ρ^2. Uniformly in q, Markov's inequality gives $\mathbb{P}_q(T > N) \leq B/N$. Take suprema and then the infimum. The original finite expected budget implies almost-sure finite stopping work; the truncated procedure terminates after at most N observations regardless. No deterministic work cap is inferred for the original procedure. □

Lemma 7.3 (Uniform hidden-magnitude rounding)**.** *For every original allocation q, define $\bar{q}_j = \pi_j^{-1} \int_{A_j} q\, d\mu$. This allocation is admissible for μ_η and preserves θ exactly. Conversely, any finite allocation $\bar{q}$ lifts to $q(d) = \bar{q}_j$ on A_j, preserving feasibility and target. For any fixed N-capped adaptive policy, the two transcript laws in either comparison have total variation at most $N\eta$. Their expected work differs by at most $N^2\eta$, and their squared risks differ by at most $\rho^2 N\eta$.*

Proof. On each fresh coin draw its original D and S jointly, and replace D by its bin representative while retaining S. The rounded coin has exactly the required mixture allocation $\bar{q}$. Its probability of heads differs from the original by at most η, even after conditioning on both hidden variables. Couple each queried pair of bits using a fresh common uniform. Use the same independent policy seed. Until the first differing bit the two policies make the same decisions, visit the same coin identities and stop together. There are at most N opportunities for a mismatch, each having conditional probability at most η. Thus the mismatch probability, and hence total variation, is at most $N\eta$. This holds under arbitrary cross-coin feedback, since the latent probability difference is pointwise bounded, not merely bounded before observations. The reverse lift uses the same coupling. Both work and squared loss are bounded respectively by N and ρ^2, and the target is unchanged, giving the two expectation bounds. Free unused identities may be removed: coins are indexed by first charged observation, so at most N genuine coin tapes are needed. □

Lemma 7.4 (A rarity-preserving finite parameter grid)**.** *For any admissible allocation $\bar{q}$ under μ_η, put $q_j^\downarrow = \gamma\lfloor \bar{q}_j/\gamma \rfloor$. Then $q^\downarrow \in Q_\gamma$, $0 \leq \theta(\bar{q}) - \theta(q^\downarrow) \leq \gamma$, and every fixed N-capped policy satisfies*

$$\mathrm{TV}(P_{\bar{q}}, P_{q^\downarrow}) \leq N\gamma, \quad |\mathbb{E}_{\bar{q}} T - \mathbb{E}_{q^\downarrow} T| \leq N^2\gamma,$$

and

$$|\mathbb{E}_{\bar{q}}(Z-\theta(\bar{q}))^2 - \mathbb{E}_{q^\downarrow}(Z-\theta(q^\downarrow))^2| \le \rho^2 N\gamma + 2\rho\gamma.$$

Proof. Rounding downward preserves the global rarity inequality, and the target change is at most $\sum_j \pi_j\gamma = \gamma$. On every new coin couple the hidden type identically and the two signs by a common uniform. Their sign disagreement probability is at most γ. If signs match, use identical bit tapes. At most N fresh coins are actually queried. Union bounding their sign disagreements gives the transcript bound uniformly over adaptive policies. The work bound follows from $0 \le T \le N$. First hold the target fixed to obtain the squared-loss change $\rho^2 N\gamma$ from total variation; then use $|(z-\theta)^2 - (z-\theta')^2| \le 2\rho|\theta-\theta'|$ for $z, \theta, \theta' \in [0,\rho]$ to account for the target shift. □

Restoring the exact expected budget. If a transferred policy has uniform expected work at most $B+D$ and squared risk at most A, independently run it with probability $p = B/(B+D)$ and otherwise output $\rho/2$ without observations. This is an allowed independent seed choice. Its work is at most B, and its squared risk is at most

$$pA + (1-p)\rho^2/4 \le A + \rho^2 D/[4(B+D)]. \tag{2}$$

This is an expected-work statement, not a high-probability or deterministic work guarantee. The extra error is explicit.

Proof of Theorem 7.1: value comparison. Start with a nearly optimal finite-grid policy at budget B. For any original q, average it within the bins and then round the averaged coordinates downward. The two preceding lemmas show that running this same policy on the original experiment has uniform squared risk at most $G_{N,\eta,\gamma}(B) + E$ and expected work at most $B + C$. Apply (2), then take the approximation error in the nearly optimal policy to zero. This gives

$$V_\mu(B) \le G_{N,\eta,\gamma}(B) + E + \rho^2 C/[4(B+C)]. \tag{3}$$

For the opposite direction start with an N-capped original policy at budget B. Every grid parameter has the bin-constant lift to the original class. Magnitude rounding alone gives squared risk increase at most $E_\eta = \rho^2 N\eta$ and work increase at most $C_\eta = N^2\eta$. Applying (2) now in the finite-grid experiment and then the truncation lemma gives

$$G_{N,\eta,\gamma}(B) \le V_\mu(B) + \rho^2 B/N + E_\eta + \rho^2 C_\eta/[4(B+C_\eta)]. \tag{4}$$

No grid approximation is needed in this direction: the grid is a subset of the feasible rounded parameters. Since $C_\eta \le C$, $E_\eta \le E$, and $D/(B+D)$ is increasing, (3)--(4) imply (1). The claimed limit follows directly from the displayed error terms. □

Exact finite-dimensional convex formulation. We give the formulation to specify what a future finite numerical solve would actually certify. A history records the ordered sequence of queried coin identities and returned bits, with identities canonically numbered by first observation. After $n < N$ bits its actions are stop, query any existing coin, or query one fresh coin. At depth N it must stop. This is a finite tree, including all feedback policies. The number of nodes may be enormous; no efficient-computation claim is made.

For each history h, let $\ell_q(h)$ be the product, over its distinct coins, of the probabilities of their local bit words under the finite mixture and grid parameter q. It excludes policy-choice probabilities. Let x_h be a nonnegative realization weight with $x_\varnothing = 1$. At a decision node, introduce action weights $x_{h,a} \ge 0$ satisfying

$$\sum_a x_{h,a} = x_h.$$

For a querying action a, both bit children obey $x_{(h,a,0)} = x_{(h,a,1)} = x_{h,a}$. Chance probabilities belong in ℓ_q, not in these policy weights. Let $t_h = x_{h,\mathrm{stop}}$ be the terminal weight. At depth N set $t_h = x_h$. Terminal probability under q is $\ell_q(h)t_h$.

Choose scaled terminal outputs y_h with $0 \le y_h \le \rho t_h$. If $t_h > 0$, output $z_h = y_h/t_h$. The uniform work constraints and squared-risk objective are the finite constraints, one pair for every $q \in Q_\gamma$,

$$\sum_h |h| \ell_q(h) t_h \le B, \qquad \sum_h \ell_q(h) \frac{(y_h - \theta(q) t_h)^2}{t_h} \le r.$$

The sums are over terminal histories. Minimize r. Each quotient is the closed quadratic perspective, assigned value zero when $t_h = y_h = 0$. The other constraints are linear, so this is a finite-dimensional convex program; continuous terminal outputs need not be discretized.

For exactness, any such action flows produce a behavioral policy by choosing action a with probability $x_{h,a}/x_h$ when $x_h > 0$; unreachable nodes may use arbitrary decisions. Conversely, arbitrary shared and local seeds do not create extra statistical strategies here. Conditional on a history of actions and bits, seed consistency contributes a parameter-independent indicator, while its observation likelihood is $\ell_q(h)$ independently of the consistent seed. Thus the conditional law of that seed, and hence of its next action, is independent of q whenever the history is possible. The resulting behavioral action probabilities have the same laws for all grid parameters. At a terminal history, replace any remaining randomized output by its conditional mean. That conditional law is again parameter independent, and Jensen's inequality decreases squared loss for every target. This proves equality with the stated policy minimax value, including shared independent randomization and arbitrary cross-coin feedback.

There is always a feasible constant-output policy. All flow and scaled-output variables range over a compact polytope; the perspectives are continuous on $0 \le y_h \le \rho t_h$ with the specified value at zero. Therefore the minimum is attained. This existence assertion does not claim that the program was solved in this research pass.

8 A concrete nonlinear feedback estimator

For a specified finite law, choose a finite prior on feasible allocations using only known μ, ρ. The prior includes the null, masses $\rho/2$ and ρ, and uniform, hardest-first, easiest-first, inverse-square and feasible single-atom profiles. The true allocation is not supplied to the policy. Conditional on each prior hypothesis, coin likelihoods factor over their observed local words. Thus an existing coin's next-bit probability can be computed from its heads/tails counts, and the posterior weights on allocations update after every actual bit.

For action a (a fresh or existing coin), let $p_a(q)$ be that next-bit probability. Its one-step posterior variance reduction is exactly

$$\frac{\mathrm{Cov}(\theta, p_a)^2}{\overline{p}_a(1 - \overline{p}_a)}, \qquad \overline{p}_a = \mathbb{E}[p_a \mid \text{observed transcript}].$$

Select the action maximizing this quantity, with a deterministic tie rule, and return the final posterior mean after exactly B flips. This is an implementable nonlinear feedback heuristic. One-step optimality under a finite prior is neither global Bayes optimality nor a guarantee over arbitrary q. No claim of calibrated uncertainty under misspecification is made.

The frozen suite uses two- and four-gap laws, $\rho = 1/4, 1/16$, five allocations including an out-of-prior mixture, budgets64,256,1024, and32 replicates per cell. Comparators use the same posterior estimator with fresh-only or fixed blocks of4 or16 flips, plus zero, half and rarity midpoint constants. All sampling procedures use the identical observation interface and exactly B charged flips. The total evaluation used 3,440,640 genuine flips in41.22seconds. The source, configurations and seeds were locked before evaluation; no post-result tuning was performed.

Feedback had a smaller MSE point estimate than the best fixed acquisition comparator in40 of60 cells, and a smaller maximum tested MSE in all12 law/rarity/budget groups. At $B = 1024$:

Law	ρ	Feedback max MSE	Best fixed max MSE
Two gaps	1/4	0.0034615	0.0044203
Two gaps	1/16	0.0005496	0.0013175
Four gaps	1/4	0.0022861	0.0054182
Four gaps	1/16	0.0004891	0.0012849

These are maxima over only five tested profiles and noisy finite-replicate estimates, not minimax certificates. Complete bias, variance, raw/clipped RMS, target-miss probability and bounded-loss uncertainty are in the data tables. The uncertainty bounds are wide and cellwise, not simultaneous. No confidence-interval estimator or coverage result is supplied. Outputs already lie in $[0, \rho]$, so clipping to $[0, 1]$ changes nothing.

Independent replay of one predesignated replicate in every one of240 cells matched outputs and acquisition counts. An independent code audit found no access to actual q, D, S and checked every stored work count. Row timings share deterministic prediction caches and omit model setup; the overall timer includes setup. There are zero extra pilot observations, but setup has nonzero computational cost. No one-shot speedup or amortized break-even claim follows from these timings.

9 Remaining proof obligations and source positioning

A successful pilot-selected production rule needs a response-weighted calibration guarantee. Conditional on a pilot, let $g = m_+ - m_-$ denote the production response and let it be calibrated at a fitted $\widehat{q}$. Its actual bias is

$$\int (q - \widehat{q})(g - 1)\, d\mu.$$

Matching a few pilot moments or an estimated target does not bound this quantity. The local-depth theorem supplies an exact failure of shallow identification, while deeper production remains a possible escape. Independent pilot and production data do not remove conditional bias.

Lee--Valiant's Section7 oracle construction optimizes stopping flows and terminal coefficients for known conditional populations and a specified cap. Learning those populations with a uniform robust calibration guarantee is an additional obligation here. Juditsky--Nemirovski's product-experiment comparison is relevant to fixed acquisition; a concurrent repository task is developing its faithful transcript transfer and is not adopted as a completed dependency. The focused primary-source audit accompanies this note. Classical moment matching, score identities and convex comparison arguments are not claimed original.

The exact feedback gain is a finite cap-two effect; the small-rarity information calculation does not amplify it into the missing logarithm. The depth-tail hard pair changes with K, so those tail bounds cannot be summed as though they held at one fixed population. The finite-prior heuristic has no uniform-q guarantee. These limitations leave the original arbitrary-law expected-budget rate and confidence dependence open.